

TASK :

1. Splash Screen
2. Login Screen (Admin / User)
3. Save login using SharedPreferences
4. Auto-login (skip login if already logged in)
5. Different screens based on role

STEP 1 : Create Project

```
cd D:\MAD_Lab
flutter create lab9_auth_app
cd lab9_auth_app
```

```
PS C:\Users\rifaq> cd D:\MAD_Lab
PS D:\MAD_Lab> flutter create lab9_auth_app
Creating project lab9_auth_app...
Resolving dependencies in `lab9_auth_app`... (4.8s)
Downloading packages...
Got dependencies in `lab9_auth_app`.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd lab9_auth_app
$ flutter run

Your application code is in lab9_auth_app\lib\main.dart.

PS D:\MAD_Lab> cd lab9_auth_app
PS D:\MAD_Lab\lab9_auth_app> █
```

STEP 2 : Install Package

```
flutter pub add shared_preferences
```

(Installing)

```
PS D:\MAD_Lab> cd lab9_auth_app
PS D:\MAD_Lab\lab9_auth_app> flutter pub add shared_preferences
Resolving dependencies...
Downloading packages... (8.6s) █
```

After Installation

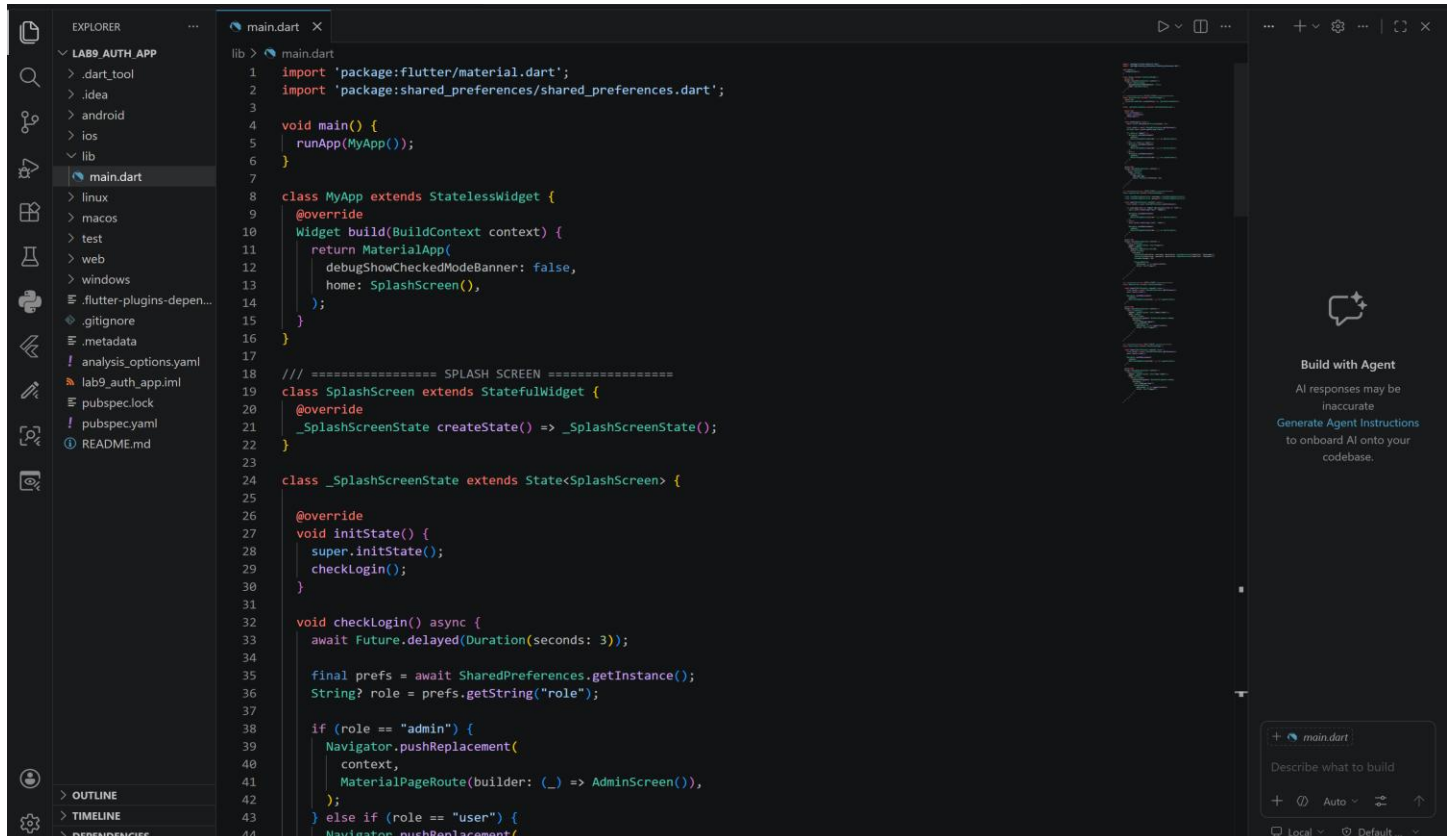
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\rifaq> cd D:\MAD_Lab
PS D:\MAD_Lab> cd lab9_auth_app
PS D:\MAD_Lab\lab9_auth_app> flutter pub add shared_preferences
"shared_preferences" is already in "dependencies". Will try to update the constraint.
Resolving dependencies...
Downloading packages...
  matcher 0.12.19 (0.12.20 available)
  meta 1.17.0 (1.18.2 available)
  test_api 0.7.10 (0.7.12 available)
  vector_math 2.2.0 (2.3.0 available)
Got dependencies!
4 packages have newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.
PS D:\MAD_Lab\lab9_auth_app> |
```

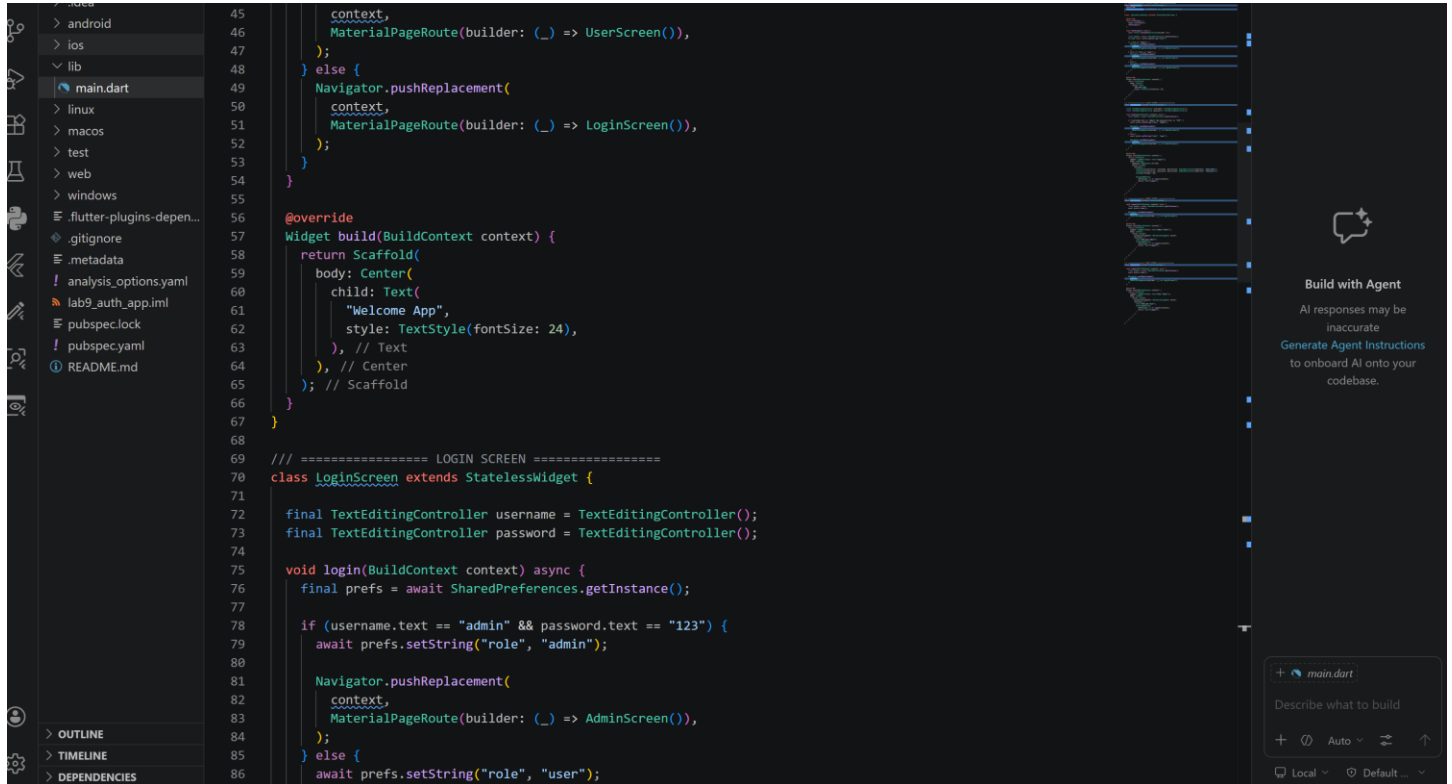
STEP 3 : Replace main.dart

lib/main.dart

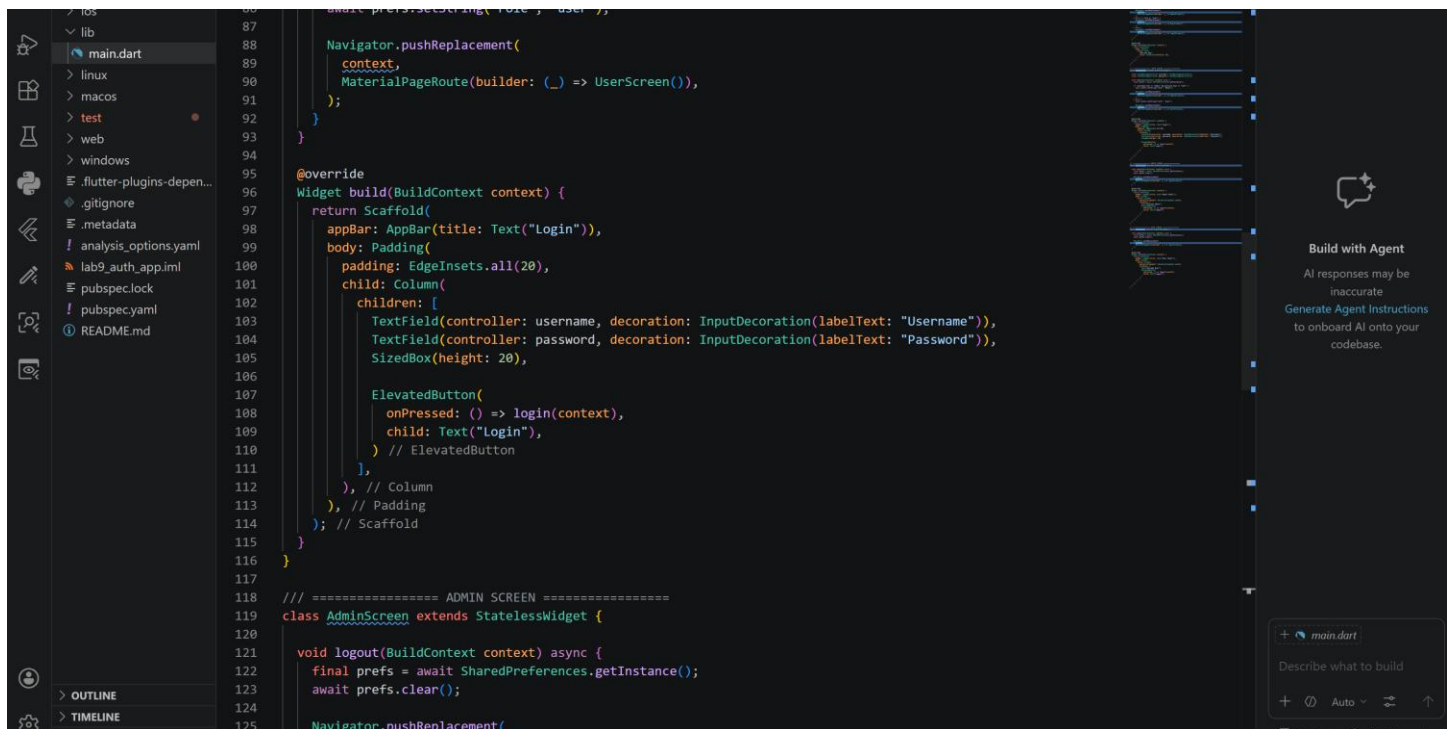
Updated Code



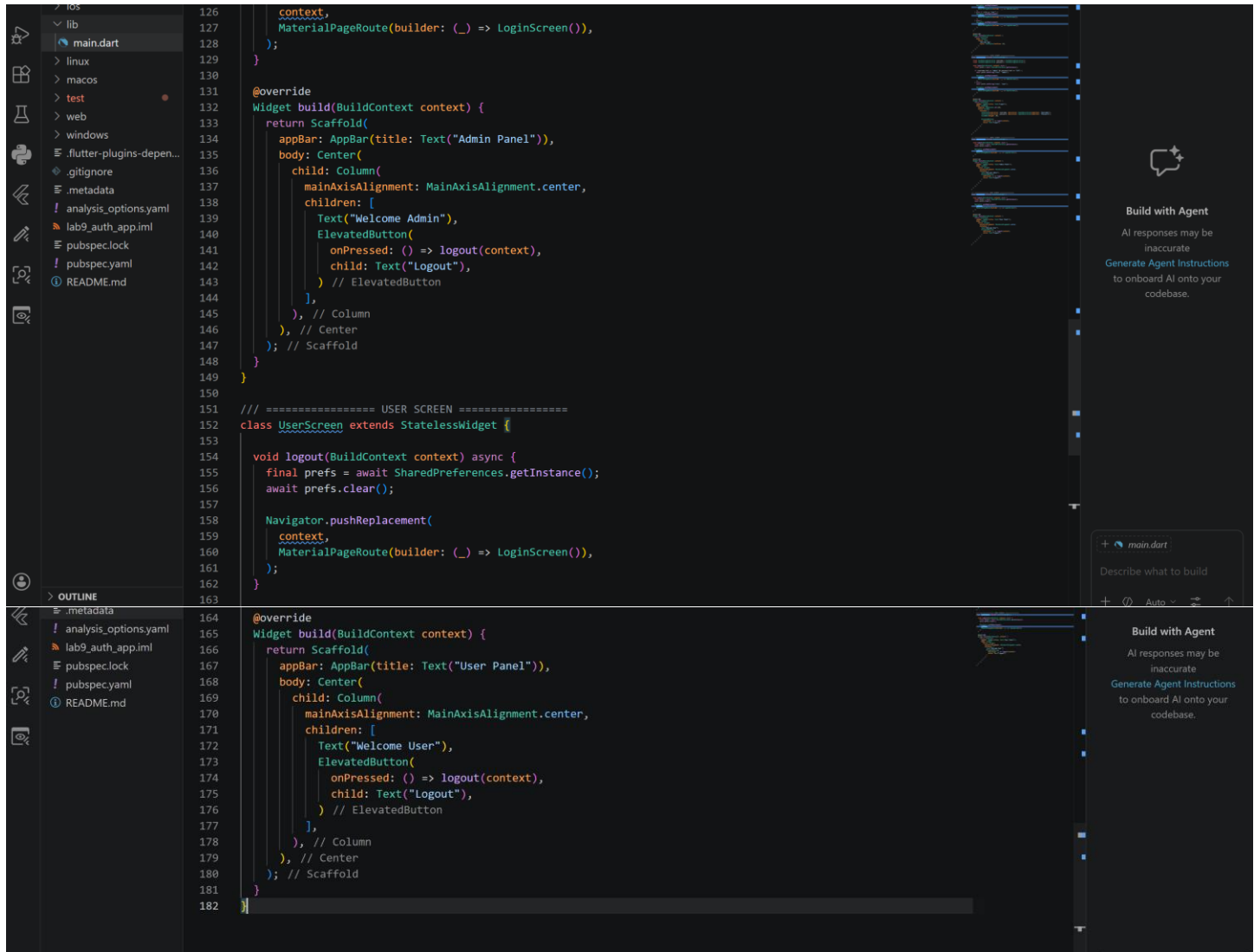
```
lib > main.dart
1 import 'package:flutter/material.dart';
2 import 'package:shared_preferences/shared_preferences.dart';
3
4 void main() {
5   runApp(MyApp());
6 }
7
8 class MyApp extends StatelessWidget {
9   @override
10  Widget build(BuildContext context) {
11    return MaterialApp(
12      debugShowCheckedModeBanner: false,
13      home: SplashScreen(),
14    );
15  }
16 }
17
18 /// ===== SPLASH SCREEN =====
19 class SplashScreen extends StatefulWidget {
20   @override
21   _SplashScreenState createState() => _SplashScreenState();
22 }
23
24 class _SplashScreenState extends State<SplashScreen> {
25
26   @override
27   void initState() {
28     super.initState();
29     checkLogin();
30   }
31
32   void checkLogin() async {
33     await Future.delayed(Duration(seconds: 3));
34
35     final prefs = await SharedPreferences.getInstance();
36     String? role = prefs.getString("role");
37
38     if (role == "admin") {
39       Navigator.pushReplacement(
40         context,
41         MaterialPageRoute(builder: (_) => AdminScreen()),
42       );
43     } else if (role == "user") {
44       Navigator.pushReplacement(
```



```
45     context,
46     MaterialPageRoute(builder: (_) => UserScreen()),
47   );
48   } else {
49     Navigator.pushReplacement(
50       context,
51       MaterialPageRoute(builder: (_) => LoginScreen()),
52     );
53   }
54 }
55
56 @override
57 Widget build(BuildContext context) {
58   return Scaffold(
59     body: Center(
60       child: Text(
61         "Welcome App",
62         style: TextStyle(fontSize: 24),
63       ), // Text
64     ), // Center
65   ); // Scaffold
66 }
67
68
69 /// ===== LOGIN SCREEN =====
70 class LoginScreen extends StatelessWidget {
71
72   final TextEditingController username = TextEditingController();
73   final TextEditingController password = TextEditingController();
74
75   void login(BuildContext context) async {
76     final prefs = await SharedPreferences.getInstance();
77
78     if (username.text == "admin" && password.text == "123") {
79       await prefs.setString("role", "admin");
80
81       Navigator.pushReplacement(
82         context,
83         MaterialPageRoute(builder: (_) => AdminScreen()),
84       );
85     } else {
86       await prefs.setString("role", "user");
```



```
87   await prefs.setString("role", "user");
88
89   Navigator.pushReplacement(
90     context,
91     MaterialPageRoute(builder: (_) => UserScreen()),
92   );
93 }
94
95 @override
96 Widget build(BuildContext context) {
97   return Scaffold(
98     appBar: AppBar(title: Text("Login")),
99     body: Padding(
100       padding: EdgeInsets.all(20),
101       child: Column(
102         children: [
103           TextField(controller: username, decoration: InputDecoration(labelText: "Username")),
104           TextField(controller: password, decoration: InputDecoration(labelText: "Password")),
105           SizedBox(height: 20),
106           ElevatedButton(
107             onPressed: () => login(context),
108             child: Text("Login"),
109           ) // ElevatedButton
110         ], // Column
111       ), // Padding
112     ); // Scaffold
113 }
114
115
116 /// ===== ADMIN SCREEN =====
117 class AdminScreen extends StatelessWidget {
118
119   void logout(BuildContext context) async {
120     final prefs = await SharedPreferences.getInstance();
121     await prefs.clear();
122
123     Navigator.pushReplacement(
```



How It Works

Action	Result
Open app	Splash screen
First time	Goes to login
Login as admin	Admin screen
Login anything else	User screen
Close & reopen	Auto login (no login screen)
Logout	Back to login

STEP 4 : Run App

```
flutter run -d chrome
```

Welcome App

Login

Username

Password

Login

User Panel

Welcome User
Logout